



### Shield & Extend the Pole Life

AEX Heat Shrink Pole Protection Sleeves are designed to fit all Pole sizes and dimensions. These Sleeves offer protection against subterranean termites, ensuring your infrastructure remains strong for decades to come.

AEX's Heat Shrink Pole Protection Sleeves, "APW" and "APT" are made from thermally stabilised Cross-linked Polyolefin material that offers abiding service reliability in severe climatic conditions and in polluted environments.

By applying heat, the Sleeve shrinks to the original smaller extruded diameter, fitting tightly over a wide range of Pole sizes. The inside-coated Bitumen mastic sealant melts and flows throughout the shrinking process and ensures a permanent, reliable, and watertight seal over even the most irregular shapes. It protects the pole from potentially hazardous environmental conditions.

### The Product Types:

- APW – Designed for the protection of Installed Poles. These are Wrap Sleeves to be folded around the Poles and then shrunk. These Sleeves are available with Bitumen mastic coating and Stainless-Steel Channels.
- APT – Designed for the protection of Un-installed / Fresh Poles against corrosion. These are Sleeves to be inserted on the fresh Poles and then shrunk. These Sleeves are available with Bitumen mastic coating.

### The causes for Utility Pole collapses:

#### ● Loss of Pole Strength due to Deterioration:

Loss of pole strength because of decay, particularly at the ground line, is a critical factor in utility infrastructure failure, as this area experiences maximum mechanical stress and holds high moisture content. Decay—both internal and external—can reduce a pole's strength from 100% to near 0% within 2-8 years.

# Heat Shrink Pole Protection Sleeves

## ● Pole damage due to Termite attacks:

Subterranean termites cause billions of dollars in damage globally each year, primarily by feeding on cellulose-rich materials like wood, and utility poles are prime targets for them. Termite damage will often go unnoticed until it is significant enough to cause pole failure, and by then it's too late!!

Over time, termites can develop resistance to common chemical treatments used for protection. Repeated exposure to the same chemicals allows termites to adapt, diminishing the effectiveness of treatments and making infestations harder to control. This resistance, combined with their ability to thrive in varied environments, makes termites a persistent and costly threat to infrastructure.

## Features & Benefits:

- Excellent electrical properties and mechanical strength.
- 100% waterproof sealing as per IP 68 against moisture and environmental contaminants
- Resistant to abrasion, weathering, atmospheric pollution and aggressive chemical agents.
- Halogen-free & UV resistant
- Infusible
- Resistant to cold flow (thermally stable)
- Quick and easy to install.
- Extended size range can accommodate Poles with almost all the dimensions.
- Easy installation on concrete, steel or composite poles.
- Use of Pole Protection Sleeves adds insignificant percentage to the total cost of New Pole replacement.
- Best & Cost-effective Solution for long-term maintenance of Poles.
- Corresponds to IEC 60684-2 Storage conditions
- Unlimited Shelf life

## Technical Properties

| Test Description     | Typical Value                    | Test Method |
|----------------------|----------------------------------|-------------|
| Tensile Strength     | 16 N/mm <sup>2</sup> (min.)      | ASTM D638   |
| Elongation           | 350% (min.)                      | ASTM D638   |
| Water Absorption     | 0.50% (max.)                     | ASTM D570   |
| Density              | 0.8 - 1.2 gm/cm <sup>3</sup>     | ASTM D792   |
| Longitudinal Change  | +/- 10 Shore D                   | ASTM D2671  |
| Hardness             | 45 +/- 10 Shore D                | ASTM D2240  |
| Thermal Ageing       |                                  |             |
| Tensile Strength     | 14 N/mm <sup>2</sup> (min.)      | ASTM D638   |
| Elongation           | 300% (min.)                      | ASTM D638   |
| Di-electric Strength | 12 kV/mm. (min.)                 | ASTM D149   |
| Volume Resistivity   | 1x10 <sup>12</sup> Ohm.cm (min.) | ASTM D257   |
| UV Resistance        | No Cracking                      | ASTM G154   |

## Sleeve for Installed Poles

| Product Dimensions |        |        |        |
|--------------------|--------|--------|--------|
| Product Code       | D Min. | d Max. | T ±10% |
| APW 55 / 8         | 55     | 8      | 2.7    |
| APW 76 / 18        | 76     | 18     | 2.7    |
| APW 105 / 28       | 105    | 28     | 2.7    |
| APW 140 / 35       | 140    | 35     | 2.7    |
| APW 190 / 46       | 190    | 46     | 2.7    |
| APW 240 / 50       | 240    | 50     | 2.7    |

D - Dia. Supplied; d - Dia. Free Recovered;  
T - Thickness after Free Recovery; All dimensions are in mm

## Sleeve for Uninstalled / Fresh Poles

| Product Dimensions |        |        |        |
|--------------------|--------|--------|--------|
| Code               | D Min. | d Max. | T ±10% |
| APT 95 / 25        | 95     | 25     | 3.0    |
| APT 115 / 34       | 115    | 34     | 3.2    |
| APT 140 / 42       | 140    | 42     | 3.2    |
| APT 180 / 55       | 180    | 55     | 3.3    |
| APT 250 / 70       | 250    | 70     | 3.5    |
| APT 320 / 90       | 320    | 90     | 3.5    |

D - Dia. Supplied; d - Dia. Free Recovered;  
T - Thickness after Free Recovery; All dimensions are in mm